LECTURE 1
INTRODUCTION TO ENERGY STORAGE REQUIREMENTS IN HEV AND EV

INTRODUCTION TO ENERGY STORAGE

Energy storage refers to a system or device that is capable of accepting energy from an external source during charging, storing that energy, and delivering the stored energy to an external load during discharge. In electric vehicles (EVs) and hybrid electric vehicles (HEVs), the energy-storage system acts as an important link between the energy source and the vehicle propulsion system. It must be capable of supplying power to the electric motor during vehicle operation and, in many applications, accepting energy recovered during regenerative braking.

The energy-storage system in an automobile is different from a stationary energy-storage system because the vehicle operates under continuously changing conditions. During acceleration, a large amount of power may be required for a short duration, whereas during cruising the power requirement may be relatively small and steady. During braking, the direction of power flow can reverse because the electric motor can operate as a generator and send energy back to the storage system. Therefore, an automotive energy-storage system must be capable of handling both energy requirements and rapidly changing power requirements.

REQUIREMENTS OF ENERGY STORAGE IN AUTOMOBILES

An automotive energy-storage system provide adequate specific energy and specific power, high efficiency, low maintenance, effective energy management, acceptable cost, environmental adaptability, environmental friendliness, and safety.

1. **Specific Energy**

   Specific energy is the amount of electrical energy stored for every kilogram of battery mass. It is given by:

   $$\text{Specific Energy} = \frac{E_{stored}}{m_b}$$

   Where E_{stored} is stored energy in Wh and m_b is battery mass in kg. The unit is Wh/kg. For example, if a battery stores 60 kWh and has a mass of 400 kg, its specific energy:

   $$\frac{60{,}000}{400} = 150 \text{ Wh/kg}$$

   Specific energy is particularly important in EVs because the mass of the battery directly affects vehicle performance. A high-specific-energy battery can store a large amount of energy without adding excessive weight to the vehicle. A lower battery mass can reduce the energy required for acceleration and rolling resistance and can contribute to improved vehicle efficiency and range. Therefore, higher specific energy is generally desirable for EV applications where long range and low vehicle mass are important. However, specific energy should not be considered alone. A battery designed for very high energy storage may not necessarily have the high-power capability required for rapid acceleration.

2. **Specific Power**

   Specific power indicates the amount of power that an energy-storage system can deliver or absorb per kilogram of battery. It is a highly variable and rather anomalous quantity,

since the power given out by the battery depends far more upon the load connected to it than the battery itself. Although batteries do have a maximum power, it is not sensible to operate them at anywhere near this maximum power for more than a few seconds, as they will not last long and would operate very inefficiently. It is expressed as:

$$\text{Specific Power} = \frac{P_b}{m_b}$$

Where P_b is battery power in W. The unit is W/kg.

For example, if a 300 kg battery can deliver 90 kW. Specific Power $= \frac{90{,}000}{300} =$ 300 W/kg.Therefore, the battery has a specific power of 300 W/kg.

Specific power is important when a vehicle requires a large amount of power for a short period. During acceleration, hill climbing, overtaking, and starting from rest, the traction motor may require a high-power output. The energy-storage system must therefore be capable of delivering this power without excessive voltage drop, overheating, or degradation.

Specific power is also important during regenerative braking because the storage system must be able to accept high charging power for a short duration. Supercapacitors are particularly suitable for this type of application because of their high power capability. Thus, specific energy relates mainly to how much energy can be stored, whereas specific power relates to how quickly energy can be delivered or absorbed.

3. **Energy efficiency**

It is defined as the ratio of electrical energy supplied by a battery to the amount of electrical energy required to return it to the state before discharge. A strong argument for using electric vehicles is based on its efficient use of energy, with resulting reduction of overall emissions; hence high energy efficiency is desirable. It should be clear from what has been said in the preceding sections that the energy efficiency will vary greatly with how a battery is used. If the battery is charged and discharged rapidly, for example, energy efficiency decreases considerably. However it does act as a guide for comparing batteries, in much the same way as fuel consumption does for cars.

The energy efficiency can be expressed as:

$$\eta_E = \frac{E_{discharged}}{E_{charged}} \times 100$$

For example, if 100 kWh is supplied to a storage system during charging and 90 kWh can later be recovered, the energy efficiency is: $\eta_E = \frac{90}{100} \times 100 = 90\%$

High efficiency is desirable because it reduces energy consumption and heat generation. This becomes especially important in EVs and HEVs because the storage system may undergo many charge-discharge cycles. Efficiency is affected by temperature, charging rate, discharging rate, SoC, internal resistance, and aging.

4. **Maintenance Requirement**

The maintenance requirement refers to the inspection, servicing, monitoring, and replacement activities required to keep the energy-storage system in proper operating condition. An automotive storage system should ideally require low maintenance because frequent maintenance increases vehicle operating costs and downtime.

Maintenance requirements may include checking electrical connections, monitoring battery temperature, inspecting cooling systems, checking insulation, diagnosing abnormal cell behavior, and verifying the operation of protection systems. Modern EV

batteries generally use electronic monitoring and control systems such as a BMS, which continuously monitors voltage, current, temperature, SoC, and other important parameters. A low-maintenance energy-storage system provides improved reliability, lower operating costs, and greater convenience for the vehicle owner.

5. **Energy Management**

Energy management refers to the control and optimization of energy flow between the energy source, energy-storage system, power electronics, and traction motor. It is essential because the power demand of a vehicle changes continuously according to speed, acceleration, road gradient, traffic conditions, and driving style.

During acceleration, the energy-storage system supplies power to the motor:

Battery → Converter → Motor

During regenerative braking, the direction of energy flow is reversed:

Motor/Generator → Converter → Battery

In an HEV, energy management also determines how power is shared between the IC engine and electric motor. The controller can determine when the engine should operate, when the battery should supply power, when regenerative braking should be used, and how the battery SoC should be maintained. Effective energy management improves vehicle efficiency, reduces unnecessary battery stress, supports regenerative braking, and can contribute to longer battery life.

6. **Cost**

Cost is an important factor in the selection of automotive energy-storage systems. The total cost is not limited to the cost of the energy-storage cells. It may also include the cost of the BMS, cooling system, enclosure, power electronics, sensors, contactors, protection devices, installation, maintenance, replacement, and recycling.

For EVs, battery cost can have a significant effect on the overall vehicle cost. Therefore, the energy-storage system should provide the required energy and power at an acceptable cost. However, the lifetime cost is more important than simply the initial purchase cost. A storage system with a higher initial cost may be economically attractive if it has a longer service life, higher efficiency, lower maintenance requirements, and better reliability. Thus, energy-storage selection involves balancing cost with performance, safety, lifetime, and environmental requirements.

7. **Environmental Adaptation and Friendliness**

An automotive energy-storage system must operate reliably under a wide range of environmental conditions. Vehicles may experience low and high temperatures, humidity, dust, vibration, mechanical shocks, rain, and rapid changes in operating conditions. The energy-storage system must continue to operate safely and provide acceptable performance under these conditions.

Temperature is particularly important. At low temperatures, battery chemical reactions can slow down, reducing available power and charging capability. At high temperatures, degradation can accelerate and safety risks can increase. Therefore, many EV battery systems use thermal-management systems to maintain cells within an appropriate temperature range.

Environmental friendliness also refers to the environmental impact of the energy-storage system throughout its life cycle. This includes raw-material extraction, manufacturing, transportation, operation, recycling, and disposal. Modern energy-

storage development therefore considers the recovery and recycling of valuable materials and the reduction of environmental impact during manufacturing and end-of-life treatment.

8. **Safety**

Safety is one of the most critical requirements for automotive energy-storage systems because they contain substantial amounts of stored electrical, chemical, or mechanical energy. The system must remain safe during normal operation as well as abnormal conditions such as overcharging, short circuits, overheating, mechanical impact, and vehicle collisions.

Electrical safety involves protection against overvoltage, overcurrent, short circuits, insulation failure, and electric shock. Fuses, contactors, isolation monitoring, and electronic protection systems can be used to protect the battery and vehicle occupants.

Thermal safety is necessary because excessive temperature can accelerate battery degradation and, for certain battery chemistries, may lead to thermal runaway. Temperature sensors, cooling systems, thermal barriers, and protective control strategies are therefore important.

Mechanical safety requires the battery enclosure and mounting structure to withstand vibration, shock, road impacts, and collision forces. The battery should remain mechanically secure and electrically isolated even under severe operating conditions.

Chemical safety involves preventing leakage or uncontrolled reactions and ensuring that the materials within the energy-storage system remain properly contained. Proper cell design, packaging, thermal management, monitoring, and protection are essential for safe operation.

BATTERY PERFORMANCE PARAMETERS

The following parameters help designers compare energy-storage technologies and select a battery suitable for a particular vehicle. Unless otherwise stated, use SI units and define symbols consistently.

1. ***Cell and battery voltages***

All electric cells have nominal voltages that gives the approximate voltage when the cell is delivering electrical power. The cells can be connected in series to give the overall voltage required. However, the voltage of a practical battery is not constant. It changes according to the state of charge, current, temperature and internal resistance. When a current is given out, the voltage will fall; when the battery is being charged, the voltage will rise.

The voltage of an individual battery cell is determined by the electrochemical reactions occurring inside that cell. Different battery chemistries therefore have different nominal cell voltages. A battery pack is constructed by connecting individual cells in series and, where necessary, in parallel. When cells are connected in series, their voltages add together. Therefore, if one cell has a nominal voltage V_c , and N_s cells are connected in series, the nominal battery voltage is approximately0

$$V_{pack,nom} = N_s V_{cell,nom}$$

For example, if each cell has a nominal voltage of 3.5 V and 80 cells are connected in series, the nominal pack voltage is approximately 280 V.

The ‘internal resistance' and the equivalent circuit of a battery is shown in **Figure 1** . The battery is represented as having a fixed voltage E (V_{oc}) , but the voltage at the

terminals is a different voltage V_t , because of the voltage across the internal resistance R_{int} . Assuming that a current I_d is flowing out of the battery, as in Fig. 1, then by basic circuit theory we can say that:

$$V_t = V_{oc} - I_d R_{int}$$

When a battery is supplying a large current, part of the voltage is lost internally because of the battery's internal resistance.

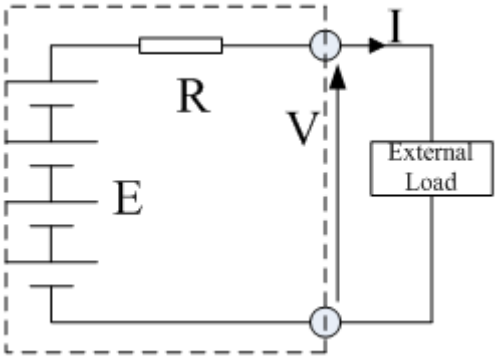


Figure 1: Simple equivalent circuit model of a battery. This battery is composed of six cells

When the discharge currentI_d is zero, the terminal voltage V_t is equal to the open-circuit voltage V_{oc}. During battery discharge, the terminal voltage decreases due to the voltage drop across the internal resistance R_{int}. Consequently, the voltage available at the battery terminals during high-current acceleration may be significantly lower than the voltage measured when the battery is at rest. In electric vehicle batteries the internal resistance should clearly be as low as possible. However, the open circuit voltage V_{oc} is not in fact constant. The voltage is also affected by the 'state of charge', and other factors such as temperature.

2. Charge Capacity (Ah)

Charge capacity is the amount of electric charge that a battery can supply during discharge before the battery reaches its specified fully discharged or cut-off condition. It represents the quantity of charge available from the active materials involved in the electrochemical reaction. During discharge, the active material at the negative electrode undergoes oxidation and releases electrons, while the positive electrode consumes these electrons through a reduction reaction. Therefore, battery capacity indicates how much electrical charge the battery can deliver, rather than directly indicating how much energy it can store.

The practical capacity of a battery is given as

$$Q_{dis} = \int_{t_0}^{t_{cut}} I_d(t)dt$$

where t_o is the time at which battery is fully charged and t_{cut} is the time at which battery terminal voltage is at V_{cut} . Therefore, $V_t\ (t_{cut}\ ) = V_{cut}$

Battery capacity is commonly expressed in ampere-hours (Ah). For constant discharge current:

$$Q_{dis}(Ah) = I_d(A) \times t(h)$$

The SI unit is the coulomb, the charge when one Amp flows for one second. However, this unit is inconveniently small. Instead, the Ah is used: one Amp flowing for one hour. The capacity of a battery might be, say, 10Ah. This means it can provide 1A for 10hrs, or 2A for 5hrs, or in theory 10A for 1hr. However, in practice, it does not work out like

this for most batteries. It is usually the case that while a battery may be able to provide 1A for 10hrs, if 10A are drawn from it, it will last less than one hour. It is most important to understand this. The capacity of the large batteries used in electric vehicles (traction batteries) is usually quoted for a 5hr discharge.

The charge capacity leads to an important notation point that should be explained at this point. The capacity of a battery in Ah is represented by the letter C. However, somewhat confusingly, until you get used to it, this is also used to represent a current. Suppose a battery has a capacity of 42 Ah, then it is said that C = 42 Ah. Battery users talk about 'a discharge current of 2C', or 'charging the battery at 0.4C'. In these cases, this would mean a discharge current of 84A, or a charging current of 16.8A

A further refinement is to give a subscript on the C symbol. As we noted above, the Ah capacity of a battery varies with the time taken for the discharge. In our example, the 42 Ah battery is rated thus for a 10hour discharge. In this more complete notation, a discharge current of 84 A should be written as $2C_{10}$.

Example: Express the current 21A from our example 42 Ah battery, in C notation. As a ratio of 42A, 21 is 1/2 or 0.5. Thus, the current $21A = 0.5C_{10}$

3. Charge/Discharge Rate

The charge/discharge rate indicates how quickly a battery is charged or discharged relative to its rated capacity. It is commonly expressed in terms of the C-rate. The C-rate relates the charging or discharging current to the rated capacity of the battery. A 1C rate means that the battery is charged or discharged at a current that would theoretically transfer its rated capacity in one hour. For example, if a battery has a rated capacity of 100 Ah, a 1C discharge rate corresponds to a current of 100 A. At this rate, the battery would theoretically be completely discharged in approximately one hour. Similarly, a 1C charging rate means that a current of 100 A is supplied for approximately one hour to charge a 100-Ah battery, subject to the battery's charging characteristics and limits.

The general relationship between C-rate, battery capacity, current, and time is

$$I = C - rate \times Q_r$$

where I is the charging or discharging current in amperes, and Q_r is the rated battery capacity in Ah. The approximate charging or discharging time can be expressed as

$$t = \frac{1}{C - rate}$$

where $t$is in hours, assuming ideal conditions and a constant current.

For a 100-Ah battery, a C/5 rate is equivalent to 0.2C. The corresponding current is

$$I = 0.2 \times 100 = 20\ A$$

so the theoretical discharge time is approximately

$$t = \frac{1}{0.2} = 5\ hrs.$$

Similarly, a 2C rate corresponds to

$$I = 2 \times 100 = 200\ A$$

and the theoretical discharge time is

$$t = \frac{1}{2} = 0.5\ hours = 30\ minutes.$$

Thus, a higher C-rate means a higher charging or discharging current and a shorter theoretical charging or discharging time.

Example for a 1-kWh battery: For a 1-kWh battery, a 0.1C rate corresponds to approximately 100 W, while a 1C rate corresponds to approximately 1 kW. Therefore, theoretically, a 1C rate would transfer an amount of energy equivalent to the rated capacity in about one hour, whereas a 0.1C rate would take approximately ten hours
For an EV, a high discharge C-rate is important during rapid acceleration, hill climbing, and other high-power conditions. A high charge C-rate is important for fast charging and for accepting high-power regenerative-braking energy.
However, the C-rate should not be interpreted as an exact real-world charging or discharging time in every situation. Actual charging time, usable capacity, heat generation, efficiency, and battery life depend on battery chemistry, temperature, SoC, internal resistance, charging strategy, and the manufacturer's specified limits.

4. ***Battery Energy***

Energy of a battery is measured in terms of the capacity and the discharge voltage. To calculate the energy, the capacity of the battery must be expressed in coulombs. 1 Ah is equivalent to 3,600 C, while 1 V refers to 1 J (J for joule) of work required to move 1 C charge from the negative to positive electrode. Therefore, the stored electrical potential energy in a 12 V, 100 Ah battery is $(12)(3.6 \times 10^5)J = 5.32\ MJ$. In general, for a nominal energy estimate:

$$E_{nom} = V_{nom} Q_r$$

where V_{nom} is the nominal no load terminal voltage. The SI unit is the Joule, but this is an inconveniently small unit, and so we use the Wh instead.

For actual energy delivered during discharge:

$$E_{out} = \int_{t_0}^{t_{cut}} V_t(t) I_d(t)\, dt \quad Wh$$

where t_o is the time at which battery is fully charged, t_{cut} is the time in hours at which battery terminal voltage is at V_{cut} , v is the battery terminal voltage and i is the battery discharge current. E_{out} is dependent on the manner in which the battery is discharged. Actual stored energy varies with discharge rate and current. However, this equation must be used with great caution. We have noted that both the battery voltage, and even more so the Ah capacity , vary considerably depending on how the battery is used. Both are reduced if the current is increased and the battery is drained quickly. The stored energy is thus a rather variable quantity, and reduces if the energy is released quickly. It is usually quoted in line with the Ah rating, i.e. if the charge capacity is given for a 5hrs discharge, then the energy should logically be given for this discharge rate.

5. **Ampere-hour (or charge) Efficiency**

$$\eta_Q = \frac{Q_{\text{discharged}}}{Q_{\text{charged}}} \times 100\%$$

In an ideal world a battery would return the entire charge put into it, in which case the amp hour efficiency is 100%. However, no battery does; its charging efficiency is less than 100%. The precise value will vary with different types of battery, temperature and rate of charge. It will also vary with the state of charge. For example, when going from about 20% to 80% charged the efficiency will usually be very close to 100%, but as the last 20% of the charge is put in the efficiency falls off greatly.

Efficiency indicates how effectively an energy-storage system can store and recover energy. During charging and discharging, some energy is lost due to internal resistance, electrochemical losses, heat generation, power-electronic losses and auxiliary systems such as cooling.

6. Energy density

Energy density is the amount of electrical energy stored per cubic metre of battery volume. It normally has units of $Wh.m^{-3}$.

$$\rho_E = \frac{E_{\text{stored}}\ (\text{Wh})}{V_b\ (\text{L})}$$

It is also an important parameter as the energy capacity of the battery (Wh) can be divided by the battery's energy density ($Wh.m^{-3}$) to show the volume of battery required. Alternatively, if a known volume is available for batteries, the volume (m^3) can be multiplied by the batteries energy density ($Wh.m^{-3}$) to give a first approximation of how much electrical energy can be made available. The battery volume may well have a considerable impact on vehicle design. As with specific energy, the energy density is a nominal figure.

7. Battery Power

The instantaneous battery terminal power is

$$p(t) = V_t\, I_d$$

where V_t is the battery terminal voltage and I_d is the battery discharge current. Using Kirchhoff's voltage law for the battery equivalent circuit

$$V_t = V_{oc} - R_{int} I_d$$

8. State of Charge (SoC)

State of Charge (SoC) is an important parameter used to describe the present charge condition of a battery. It indicates the amount of charge currently available in the battery relative to its defined available or rated capacity and is generally expressed as a percentage. In simple terms, SoC tells us how much charge is remaining in the battery at a particular instant. An SoC of 100% represents a fully charged battery within its specified charging limits, whereas an SoC of 0% represents the specified fully discharged condition. Therefore, SoC provides an indication of the remaining charge available for operating an electric vehicle or hybrid electric vehicle.

Mathematically, SoC can be expressed as the ratio of the charge currently available in the battery to its reference or usable capacity. It can be written as

$$SoC(t) = \frac{Q_{\text{available}}(t)}{Q_r} \times 100$$

where $Q_{\text{available}}(t)$is the charge available in the battery at time t, and Q_r is the rated or defined usable capacity of the battery. If a battery has a usable capacity of 100 Ah and 60 Ah of charge is currently available, then its SoC is

$$SOC = \frac{60}{100} \times 100 = 60\%$$

Thus, the battery is operating at approximately 60% SoC.

So

C can also be determined by measuring the battery current over time. This method is known as Coulomb counting. The current is the rate of change of charge given by

$$i\,(t) = \frac{dq}{dt}$$

where q is the charge moving through the circuit. The instantaneous theoretical state of charge $SoC_T\,(t)$ is the amount of equivalent charge remaining at the positive electrode and ready to be released by the energized material. If the state of charge is Q_T at the initial time t_o , then $SoC_T\,(t_o\,) = Q_T$. For a time interval dt,

$$dSoC_T = -dq = -i\,(t)dt$$

Integrating from the initial time t_o to the final time t, the expression for instantaneous state of charge is obtained as

$$SoC_T(t) = Q_r - \int_{t_0}^{t} i(\tau)\; i\, d\tau$$

The state of charge is often expressed as a percentage of the capacity of the battery as follows:

$$SoC_T(t) = \frac{Q_r - \int_{t_0}^{t} i(\tau)\; i\, d\tau}{Q_r} \times 100\%$$

The state of charge will be increasing when a battery is being charged. If the state of charge is zero initially at t = 0, the state of charge at time t expressed in percentage form is given by

$$SoC_T(t) = \frac{\int_{t_0}^{t} i(\tau)\; i\, d\tau}{Q_r} \times 100\%$$

The sign convention may change depending on whether charging or discharging current is considered positive. During charging, electrical energy enters the battery and SoC increases, whereas during discharging, the battery supplies electrical energy to the load and SoC decreases. In an EV, for example, when the battery supplies power to the traction motor, the SoC gradually decreases. During regenerative braking, the traction motor operates as a generator and sends electrical energy back to the battery, causing the SoC to increase.

9. **<u>State of Discharge (SoD)</u>**

State of Discharge (SoD) is a measure of the amount of electric charge that has been drawn from a battery relative to its total available capacity. It indicates how much of the battery's capacity has already been used during discharge. In simple terms, SoD tells us how much charge has been consumed from the battery, whereas State of Charge (SoC) tells us how much charge remains available. Therefore, SoD is an important parameter for understanding the discharge condition of a battery and is particularly useful in the analysis and modelling of batteries used in EVs and HEVs.

Mathematically, the charge removed from a battery during discharge can be calculated by integrating the battery current over time. If $i(\tau)$is the discharge current, the state of discharge can be expressed as

$$SoD_T(t) = \Delta q = \int_{t_0}^{t} i\,(\tau)d\tau$$

where $SoD(t)$represents the amount of charge removed from the battery, $i(\tau)$is the battery discharge current, $t_0$is the initial time, and $t$is the time at which the SoD is being calculated.

10. **Depth of Discharge (DoD)**

The depth of discharge (DoD) is a measure of how much of a battery's available or rated capacity has been used or discharged. It is expressed as a percentage of the battery's total reference capacity. In simple terms, DoD tells us how deeply the battery has been discharged from its fully charged condition. A DoD of 0% means that no capacity has been used and the battery is considered fully charged, while a DoD of 100% means that the entire specified usable capacity has been discharged. Therefore, DoD is an important parameter for evaluating battery operation, especially in EVs and HEVs, because repeated deep discharge can influence battery aging, cycle life, and available energy.

The mathematical expression for depth of discharge i

$$DoD(t) = \frac{Q_r - SoC_T(t)}{Q_r} \times 100\%$$

where $Q_T$is the total rated or reference capacity and $SoC_T(t)$represents the charge remaining in the battery at time t. Since the amount of charge removed from the battery can be obtained by integrating the discharge current, DoD can also be expressed as

$$DoD(t) = \frac{\int_{t_0}^{t} i\,(\tau)d\tau}{Q_r} \times 100\%$$

where $i(\tau)$is the battery discharge current, $t_0$is the initial time, and $Q_T$is the rated capacity. The integral $\int i\,(\tau)d\tau$represents the total amount of charge removed from the battery during the discharge period.

DoD and SoC are closely related. Under the same reference-capacity definition,

$$DoD(\%) = 100 - SoC(\%)$$

For example, if an EV battery is operating at 80% SoC, then its corresponding DoD is approximately $DoD = 100 - 80 = 20\%$. This means that approximately 20% of the reference capacity has been used. If the battery reaches 50% SoC, the corresponding DoD is 50%. Similarly, a battery at 20% SoC has a DoD of 80%.

11. Self-discharge rates

Most batteries discharge when left unused, and this is known as self-discharge. This is important as it means some batteries cannot be left for long periods without recharging. The rate varies with battery type, and with other factors such as temperature; higher temperatures greatly increase self-discharge.

12. Battery temperature, heating and cooling needs

Although most batteries run at ambient temperature, some run at higher temperatures and need heating to start with and then cooling when in use. In others, battery performance drops off at low temperatures, which is undesirable, but this problem could be overcome by heating the battery. When choosing a battery the designer needs to be aware of battery temperature, heating and cooling needs, and has to take these into consideration during the vehicle design process.

13. Battery life and number of deep cycles

Most rechargeable batteries will only undergo a few hundred deep cycles to 20% of the battery charge. However, the exact number depends on the battery type, and also on the details of the battery design, and on how the battery is used. This is a very important figure in a battery specification, as it reflects in the lifetime of the battery, which in turn reflects in electric vehicle running costs.